

Linear Approximation and L'Hôpital's Rule



Tangent line approximation

- 1 Use linear approximation to estimate $\sqrt{4.1}$, given $f(x) = \sqrt{x}$, $f(4) = 2$, and $f'(4) = \frac{1}{4}$.
- 2 Use linear approximation to estimate $(1.02)^5$, using $f(x) = x^5$ near $x = 1$.
- 3 Find the linearization $L(x)$ of $f(x) = \sin x$ at $x = 0$, and use it to approximate $\sin(0.1)$.

Error and over/underestimation

- 4 For $f(x) = \sqrt{x}$ (concave down for $x > 0$), explain whether the linear approximation at $x = 4$ overestimates or underestimates $f(4.1)$.
- 5 For $f(x) = x^2$ (concave up everywhere), explain whether the tangent line approximation at any point overestimates or underestimates nearby function values.

Indeterminate forms and L'Hôpital's Rule

- 6 Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ using L'Hôpital's Rule, and confirm it matches the known limit.
- 7 Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$ using L'Hôpital's Rule (applied twice).
- 8 Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$ using L'Hôpital's Rule.

Recognizing when L'Hôpital's Rule applies

- 9 Explain why L'Hôpital's Rule cannot be directly applied to $\lim_{x \rightarrow 0} \frac{x+1}{x}$, and evaluate the limit by other means.

Differentials

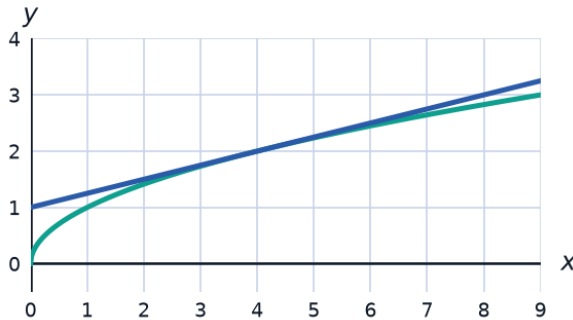
- 10 For $f(x) = x^3$, find the differential dy when $x = 2$ and $dx = 0.1$, and compare it to the actual change $\Delta y = f(2.1) - f(2)$.

More L'Hôpital's Rule practice

- 11 Evaluate $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$ using L'Hôpital's Rule (applied as many times as needed).

Visualizing a linear approximation

- 12 The graph shows $f(x) = \sqrt{x}$ together with its tangent line at $x = 4$, $L(x) = 2 + \frac{1}{4}(x - 4)$.



L'Hôpital's Rule with exponential indeterminate forms

- 13 Evaluate $\lim_{x \rightarrow \infty} \frac{x^2}{e^x}$ using L'Hôpital's Rule (applied twice).

Linear approximation in an applied context

- 14 The side length of a cube is measured as 5 cm, with a possible measurement error of ± 0.02 cm. Use the differential $dV = 3s^2 ds$ to estimate the maximum resulting error in the computed volume $V = s^3$.

A capstone L'Hôpital's Rule problem

- 15 Evaluate $\lim_{x \rightarrow 0^+} x \ln x$ by first rewriting it as a quotient $\frac{\ln x}{1/x}$ to apply L'Hôpital's Rule.
- 16 A surveyor measures the angle of elevation to a tower as $\theta = 60^\circ$ with a possible error of $\pm 1^\circ$ ($\pm \frac{\pi}{180}$ radians). If the tower's height is estimated using $h = 100 \tan \theta$ (with the base 100 m away), use the differential $dh = 100 \sec^2 \theta d\theta$ to estimate the resulting error in the height, to two decimal places.
- 17 Evaluate $\lim_{x \rightarrow \infty} x \sin\left(\frac{1}{x}\right)$ by rewriting it as $\frac{\sin(1/x)}{1/x}$ and applying L'Hôpital's Rule (or recognizing the standard limit form).

- 18 For $f(x) = \cos x$ near $x = 0$, complete each step:
- a Find the linearization $L(x)$ at $x = 0$.
 - b Use $L(x)$ to estimate $\cos(0.05)$.
 - c Compare to the true value $\cos(0.05) \approx 0.99875$, and state whether the linearization over- or under-estimates.
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Multiple choice — easy 15

- 19 Use linear approximation to estimate $f(2.9)$, given $f(3) = 2$ and $f'(3) = -1$.
- a. 2.1
 - b. 2
 - c. 1.9
 - d. 2.2
- 20 Use linear approximation to estimate $f(4.2)$, given $f(4) = -3$ and $f'(4) = -3$.
- a. -2.4
 - b. -4.2
 - c. -3
 - d. -3.6
- 21 Use linear approximation to estimate $f(1.3)$, given $f(1) = -2$ and $f'(1) = -9$.
- a. 0.7
 - b. -7.4
 - c. -2
 - d. -4.7

- 22 Use linear approximation to estimate $f(2.2)$, given $f(2) = 3$ and $f'(2) = -8$.
- a. 4.6
 - b. 1.4
 - c. -0.2
 - d. 3
- 23 Use linear approximation to estimate $f(1.9)$, given $f(2) = -5$ and $f'(2) = 9$.
- a. -5.9
 - b. -6.8
 - c. -5
 - d. -4.1
- 24 Use linear approximation to estimate $f(1.3)$, given $f(1) = 3$ and $f'(1) = -8$.
- a. 0.6
 - b. 3
 - c. 5.4
 - d. -1.8
- 25 Use linear approximation to estimate $f(6.1)$, given $f(6) = 8$ and $f'(6) = 7$.
- a. 7.3
 - b. 8.7
 - c. 9.4
 - d. 8

- 26 Use linear approximation to estimate $f(3.3)$, given $f(3) = 7$ and $f'(3) = -7$.
- a. 2.8
 - b. 4.9
 - c. 9.1
 - d. 7
- 27 Use linear approximation to estimate $f(5.5)$, given $f(6) = -1$ and $f'(6) = 1$.
- a. -1.5
 - b. -2
 - c. -1
 - d. -0.5
- 28 Use linear approximation to estimate $f(2.3)$, given $f(2) = -4$ and $f'(2) = 3$.
- a. -4.9
 - b. -3.1
 - c. -2.2
 - d. -4
- 29 Use linear approximation to estimate $f(1.5)$, given $f(1) = 6$ and $f'(1) = 5$.
- a. 6
 - b. 3.5
 - c. 11
 - d. 8.5

- 30 Use linear approximation to estimate $f(6.2)$, given $f(6) = 5$ and $f'(6) = 4$.
- a. 6.6
 - b. 5.8
 - c. 4.2
 - d. 5
- 31 Use linear approximation to estimate $f(7.5)$, given $f(8) = 7$ and $f'(8) = -9$.
- a. 11.5
 - b. 7
 - c. 16
 - d. 2.5
- 32 Use linear approximation to estimate $f(7.8)$, given $f(8) = -2$ and $f'(8) = 3$.
- a. -2.6
 - b. -3.2
 - c. -2
 - d. -1.4
- 33 Use linear approximation to estimate $f(4.8)$, given $f(5) = 6$ and $f'(5) = -9$.
- a. 9.6
 - b. 4.2
 - c. 6
 - d. 7.8

Multiple choice — medium 15

34 For $f(x) = 4x^3$, find the differential dy when $x = 4$ and $dx = 0.1$.

- a. 38.4
- b. 6.4
- c. 19.2
- d. -19.2

35 For $f(x) = x^3$, find the differential dy when $x = 6$ and $dx = 0.3$.

- a. 64.8
- b. -32.4
- c. 32.4
- d. 10.8

36 For $f(x) = x^3$, find the differential dy when $x = 4$ and $dx = -0.5$.

- a. -24
- b. -8
- c. -48
- d. 24

37 For $f(x) = 2x^2$, find the differential dy when $x = 5$ and $dx = -0.1$.

- a. -2
- b. -1
- c. 2
- d. -4

38 For $f(x) = 6x^2$, find the differential dy when $x = 6$ and $dx = -0.1$.

- a. -3.6
- b. 7.2
- c. -7.2
- d. -14.4

39 For $f(x) = 4x^3$, find the differential dy when $x = 4$ and $dx = -0.2$.

- a. -38.4
- b. -12.8
- c. -76.8
- d. 38.4

40 For $f(x) = 5x^3$, find the differential dy when $x = 2$ and $dx = 0.2$.

- a. 12
- b. 4
- c. 24
- d. -12

41 For $f(x) = 5x^2$, find the differential dy when $x = 3$ and $dx = 0.3$.

- a. 18
- b. -9
- c. 4.5
- d. 9

- 42 For $f(x) = 6x^3$, find the differential dy when $x = 6$ and $dx = -0.3$.
- a. -64.8
 - b. 194.4
 - c. -194.4
 - d. -388.8
- 43 For $f(x) = 4x^3$, find the differential dy when $x = 2$ and $dx = 0.5$.
- a. 8
 - b. -24
 - c. 48
 - d. 24
- 44 For $f(x) = 6x^3$, find the differential dy when $x = 2$ and $dx = -0.1$.
- a. -14.4
 - b. -2.4
 - c. -7.2
 - d. 7.2
- 45 For $f(x) = 2x^2$, find the differential dy when $x = 4$ and $dx = 0.2$.
- a. 3.2
 - b. -3.2
 - c. 1.6
 - d. 6.4

46 For $f(x) = 4x^3$, find the differential dy when $x = 6$ and $dx = -0.5$.

- a. 216
- b. -72
- c. -216
- d. -432

47 For $f(x) = 3x^2$, find the differential dy when $x = 1$ and $dx = 0.2$.

- a. 1.2
- b. 2.4
- c. -1.2
- d. 0.6

48 For $f(x) = 5x^3$, find the differential dy when $x = 1$ and $dx = -0.5$.

- a. -2.5
- b. 7.5
- c. -15
- d. -7.5

Multiple choice — hard 15

49 Evaluate $\lim_{x \rightarrow 0} \frac{e^{8x} - 1}{x}$ using L'Hôpital's Rule.

- a. -8
- b. 8
- c. 4
- d. $\frac{1}{8}$

50 Evaluate $\lim_{x \rightarrow 0} \frac{e^{4x} - 1}{3x}$ using L'Hôpital's Rule.

- a. $\frac{4}{3}$
- b. $\frac{3}{4}$
- c. 1
- d. $-\frac{4}{3}$

51 Evaluate $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{2x}$ using L'Hôpital's Rule.

- a. 1
- b. $-\frac{3}{2}$
- c. $\frac{2}{3}$
- d. $\frac{3}{2}$

52 Evaluate $\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x}$ using L'Hôpital's Rule.

- a. 1
- b. $-\frac{1}{2}$
- c. $-\frac{3}{2}$
- d. -1

53 Evaluate $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{5x}$ using L'Hôpital's Rule.

- a. $\frac{1}{2}$
- b. $\frac{3}{5}$
- c. $-\frac{3}{5}$
- d. $\frac{5}{3}$

54 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^2 + 3x + 4}{8x^2 + 7x}$ using L'Hôpital's Rule, applied twice.

a. $\frac{3}{4}$

b. $\frac{5}{9}$

c. $\frac{8}{5}$

d. $\frac{5}{8}$

55 Evaluate $\lim_{x \rightarrow \infty} \frac{8x^2 - 6x - 5}{7x^2 - 2x + 8}$ using L'Hôpital's Rule, applied twice.

a. $\frac{8}{7}$

b. $\frac{9}{7}$

c. $\frac{7}{8}$

d. 1

56 Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 - 5}{7x^2 + x + 9}$ using L'Hôpital's Rule, applied twice.

a. $\frac{7}{2}$

b. $\frac{1}{4}$

c. $\frac{3}{7}$

d. $\frac{2}{7}$

57 Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 + 3x - 7}{5x^2 + 2x + 3}$ using L'Hôpital's Rule, applied twice.

a. $\frac{3}{5}$

b. $\frac{5}{2}$

c. $\frac{1}{3}$

d. $\frac{2}{5}$

58 Evaluate $\lim_{x \rightarrow \infty} \frac{7x^2 - 6x - 9}{3x^2 - 7x + 8}$ using L'Hôpital's Rule, applied twice.

a. $\frac{7}{4}$

b. $\frac{8}{3}$

c. $\frac{7}{3}$

d. $\frac{3}{7}$

59 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(8x)}{x^2}$ using L'Hôpital's Rule, applied twice.

a. -32

b. 4

c. 16

d. 32

60 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x^2}$ using L'Hôpital's Rule, applied twice.

a. 2

b. 1

c. -2

d. 3

61 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(6x)}{x^2}$ using L'Hôpital's Rule, applied twice.

a. 18

b. 3

c. 9

d. -18

62 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2}$ using L'Hôpital's Rule, applied twice.

a. $\frac{1}{2}$

b. 0

c. $-\frac{1}{2}$

d. $\frac{1}{4}$

63 Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(5x)}{x^2}$ using L'Hôpital's Rule, applied twice.

a. $\frac{5}{2}$

b. $\frac{25}{4}$

c. $-\frac{25}{2}$

d. $\frac{25}{2}$